

Boundary Compensation Origin III: Non-Factorizable Holonomy and Frustrated Multi-Shadow Spectral Geometry Edge Signs, Cycle Products, Gauge Classes and Horizon Diagnostics in Structural Shadow Operators

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Version v0.1.0-manuscript draft, June 2026

Abstract

BC-Origin I introduced a minimal constructive toy model in which observable shadows arise from hidden oriented residual generators. In that model a hidden index-winding pair $h = (q, n)$ produces a scale channel $|n|$, an orientation channel $\text{sgn}(n)$, a structural scale denominator, and a signed two-shadow spectral operator. BC-Origin II replaced the free pairwise coupling constant κ by a structural kernel $K_{ij} = K(n_i, n_j)$ on hidden winding-index space. It also identified a limitation of the baseline sign rule: the signs $s_{ij} = \text{sgn}(n_i n_j)$ are vertex-factorizable and therefore balanced on every cycle. Consequently, BC-Origin II does not produce genuine cycle-level frustration without additional non-factorizable edge data.

The present paper develops the next finite-dimensional toy-model layer. We introduce a symmetric edge-holonomy field $\varepsilon_{ij} \in \{-1, +1\}$ on the coupling graph and distinguish vertex-factorizable signs from non-factorizable cycle data. Under vertex reorientation, $\varepsilon_{ij} \mapsto g_i \varepsilon_{ij} g_j$, while the cycle product

$$\Omega_C = \prod_{(ij) \in C} \varepsilon_{ij}$$

is invariant. A sign field is balanced precisely when all cycle products are $+1$; a cycle with $\Omega_C = -1$ carries a non-removable $\mathbb{Z}_2$ holonomy and is frustrated in the finite-dimensional graph sense used here.

The baseline BC-Origin III operator is the holonomy-weighted off-diagonal multi-shadow operator

$$(D_N^{\text{hol}})_{ii} = d_i, \quad (D_N^{\text{hol}})_{ij} = \eta \chi_{ij} K_{ij}, \quad i \neq j,$$

where $\chi_{ij} = \sigma_i \varepsilon_{ij} \sigma_j$ and $\sigma_i = \text{sgn}(n_i)$. This convention keeps the core spectral statements gauge-clean: vertex reorientation changes the matrix by orthogonal conjugation and hence preserves its spectrum. For an identical three-shadow complete graph with edge magnitude k , the balanced triangle has spectral type

$$d + 2\eta k, \quad d - \eta k, \quad d - \eta k,$$

whereas a frustrated triangle has

$$d + \eta k, \quad d + \eta k, \quad d - 2\eta k.$$

Thus non-factorizable holonomy changes the spectral class of the multi-shadow operator and changes the admissibility-horizon margins of the toy shadows.

The construction is a finite-dimensional structural model. It is not a physical lattice gauge theory, not a spin-glass model of matter, not a derivation of particle physics, not a theory of physical time, and not an empirical claim. Its controlled contribution is to show how the structural-kernel shadow geometry of BC-Origin II acquires genuine cycle-level obstruction once non-factorizable edge holonomy is admitted.

Keywords. Boundary Compensation; BC-Origin; structural kernels; edge holonomy; graph frustration; finite-dimensional spectral geometry; hidden winding index; admissibility horizon; toy model; gauge equivalence.

Software repository. The accompanying computational companion is intended for the public repository <https://github.com/AIDevelopersMonster/boundary-compensation>.

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1 Introduction

The BC-Origin branch asks whether observable elementary structures can be modeled as non-fundamental shadows of hidden residual generators. The first two modules established the minimal static architecture.

BC-Origin I introduced the hidden generator

$$h = (q, n), \quad n \in \mathbb{Z} \setminus \{0\}, \quad (1)$$

with two observable channels:

$$n \mapsto (|n|, \text{sgn}(n)). \quad (2)$$

The scale channel enters a closure denominator

$$d_i = 2\pi|n_i| - \theta(|q_i|), \quad (3)$$

and the orientation channel enters a signed two-shadow operator. In that minimal model, observable scale ratios are interpreted as

$$S_a = \frac{\mu_{\text{eff}}}{\lambda_a}, \quad \lambda_a > 0, \quad (4)$$

where λ_a is an eigen-denominator. The admissibility horizon is the condition $\lambda_a = 0$: as $\lambda_a \rightarrow 0^+$ the scale ratio diverges and the branch no longer defines a localized shadow inside the toy model.

BC-Origin II addressed the main weakness of the first model. The off-diagonal coupling κ should not be chosen independently for every pair of shadows. The central replacement was

$$\kappa \longrightarrow \kappa_{ij} = K(n_i, n_j), \quad (5)$$

where K is a declared structural kernel on hidden winding-index space. This replacement converts pairwise numerical fitting into a structural rule.

However, BC-Origin II also revealed a new limitation. Its baseline sign rule

$$s_{ij} = \text{sgn}(n_i n_j) = \sigma_i \sigma_j, \quad \sigma_i := \text{sgn}(n_i), \quad (6)$$

is vertex-factorizable. Every cycle product is $+1$. Therefore the baseline graph is balanced and cannot produce genuine cycle-level frustration. The present paper closes exactly this gap.

The new object is an edge-holonomy field

$$\varepsilon_{ij} = \varepsilon_{ji} \in \{-1, +1\}, \quad (7)$$

which is not assumed to be expressible as a product of vertex signs. This is the minimal additional structure needed to make the multi-shadow graph carry a non-removable cycle obstruction.

2 Scope and non-scope

The paper is deliberately narrow. It studies finite-dimensional real symmetric toy operators on a finite graph of shadows. Words such as “holonomy”, “gauge”, and “frustration” are used in their graph-theoretic and finite-dimensional toy-model sense.

The paper does not claim:

- to construct physical gauge fields;
- to derive a real spin-glass material;
- to produce particle masses or Standard-Model data;
- to define physical time or dynamics;
- to validate the BC-Origin program empirically.

The positive claim is smaller and sharper: non-factorizable edge holonomy is the minimal graph-level extension of BC-Origin II required for genuine cycle obstruction in the structural multi-shadow operator.

3 Hidden winding labels, kernels and baseline signs

Let

$$I_N = \{n_1, \dots, n_N\} \subset \mathbb{Z} \setminus \{0\} \quad (8)$$

be a finite set of hidden winding labels. A declared kernel gives a symmetric matrix

$$K_{ij} = K(n_i, n_j), \quad K_{ij} = K_{ji}, \quad K_{ii} = 1. \quad (9)$$

Examples include the index-space Lorentzian kernel

$$K_{ij} = \frac{1}{1 + (n_i - n_j)^2} \quad (10)$$

and the exponential kernel

$$K_{ij} = e^{-\alpha|n_i - n_j|}. \quad (11)$$

These are declared index-space kernels unless an independent hidden geometry is supplied. In particular, the Lorentzian expression should not be confused with an off-diagonal Fourier matrix element of $(1 - \Delta_{S^1})^{-1}$ between pure Fourier modes.

The baseline orientation signs are

$$\sigma_i = \text{sgn}(n_i), \quad s_{ij} = \sigma_i \sigma_j. \quad (12)$$

For any cycle $C = (i_1 i_2 \cdots i_m i_1)$,

$$\prod_{r=1}^m s_{i_r i_{r+1}} = \prod_{r=1}^m \sigma_{i_r} \sigma_{i_{r+1}} = \prod_{r=1}^m \sigma_{i_r}^2 = +1. \quad (13)$$

Thus the baseline BC-Origin II sign graph is always balanced.

4 Edge holonomy and vertex gauge

Let $G = (V, E)$ be an undirected finite graph whose vertices represent shadows. Edges represent declared structural couplings.

Definition 1 (Edge holonomy field). *A $\mathbb{Z}_2$ edge-holonomy field on G is a symmetric assignment*

$$\varepsilon_{ij} = \varepsilon_{ji} \in \{-1, +1\}, \quad (ij) \in E. \quad (14)$$

A vertex reorientation is a choice

$$g_i \in \{-1, +1\}, \quad i \in V. \quad (15)$$

It acts on edge signs by

$$\varepsilon_{ij} \mapsto \varepsilon'_{ij} = g_i \varepsilon_{ij} g_j. \quad (16)$$

This transformation is a finite graph analogue of a gauge change. It changes local edge representatives but not the cycle obstruction.

Definition 2 (Cycle product). *For a cycle $C = (i_1 i_2 \cdots i_m i_1)$, define*

$$\Omega_C(\varepsilon) = \prod_{r=1}^m \varepsilon_{i_r i_{r+1}}. \quad (17)$$

The cycle is balanced if $\Omega_C = +1$ and frustrated if $\Omega_C = -1$.

Proposition 1 (Gauge invariance of cycle products). *For every vertex reorientation $g_i \in \{-1, +1\}$,*

$$\Omega_C(\varepsilon') = \Omega_C(\varepsilon). \quad (18)$$

Proof. Substituting $\varepsilon'_{ij} = g_i \varepsilon_{ij} g_j$ gives

$$\prod_{r=1}^m g_{i_r} \varepsilon_{i_r i_{r+1}} g_{i_{r+1}} = \left(\prod_{r=1}^m \varepsilon_{i_r i_{r+1}} \right) \left(\prod_{r=1}^m g_{i_r}^2 \right) = \Omega_C(\varepsilon), \quad (19)$$

because $g_i^2 = 1$. □

Theorem 1 (Balanced sign fields). *A $\mathbb{Z}_2$ edge field on a connected graph is vertex-factorizable,*

$$\varepsilon_{ij} = h_i h_j \quad (20)$$

for some vertex signs $h_i \in \{-1, +1\}$, if and only if all cycle products are $+1$.

Proof. If $\varepsilon_{ij} = h_i h_j$, then the product around any cycle telescopes to $\prod_i h_i^2 = 1$.

Conversely, choose a root vertex r and set $h_r = 1$. For any vertex v , define h_v as the product of edge signs along any path from r to v . The all-cycle-positive condition makes this path-independent: two different paths form a closed cycle after reversing one path, whose product is $+1$. Therefore $h_i h_j = \varepsilon_{ij}$ for every edge. $\square$

This theorem is the formal reason BC-Origin II could not generate genuine frustration using only $s_{ij} = \sigma_i \sigma_j$.

5 Holonomy-weighted multi-shadow operator

The complete sign entering the holonomy-weighted operator is

$$\chi_{ij} = \sigma_i \varepsilon_{ij} \sigma_j. \quad (21)$$

Because the baseline $\sigma_i \sigma_j$ is itself vertex-factorizable, the non-removable cycle information is carried entirely by ε .

The preferred BC-Origin III baseline is the gauge-clean off-diagonal operator

$$(D_N^{\text{hol}})_{ii} = d_i, \quad (22)$$

$$(D_N^{\text{hol}})_{ij} = \eta \chi_{ij} K_{ij}, \quad i \neq j. \quad (23)$$

The adjective “gauge-clean” means the following. If vertex signs are changed by g_i , define the diagonal orthogonal matrix

$$G = \text{diag}(g_1, \dots, g_N). \quad (24)$$

Then the transformed operator is

$$D' = G D G. \quad (25)$$

Therefore

$$\text{spec}(D') = \text{spec}(D). \quad (26)$$

Only gauge-invariant cycle data can affect the spectral class.

Remark 1 (Diagonal signed-shift variant). *One can also study a convention-dependent diagonal-shift variant*

$$(D_N^{\text{diag}})_{ii} = d_i + \gamma \sum_{j \neq i} \chi_{ij} K_{ij}. \quad (27)$$

This variant is useful as a diagnostic model, but row sums of edge signs are not gauge-invariant. The baseline spectral theorems in this paper therefore use the off-diagonal holonomy operator.

6 Minimal triangle example

The smallest graph with possible frustration is a triangle. Take three identical shadows:

$$d_1 = d_2 = d_3 = d, \quad (28)$$

and equal edge magnitude

$$K_{ij} = k, \text{ for } i \neq j. \quad (29)$$

Absorb ηk into a single coupling scale.

6.1 Balanced triangle

For the balanced representative $\varepsilon_{12} = \varepsilon_{23} = \varepsilon_{13} = +1$, the operator is

$$D_{\text{bal}} = dI + \eta k \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}. \quad (30)$$

Its spectrum is

$$\text{spec}(D_{\text{bal}}) = \{d + 2\eta k, d - \eta k, d - \eta k\}. \quad (31)$$

6.2 Frustrated triangle

For a frustrated representative, choose

$$\varepsilon_{12} = +1, \quad \varepsilon_{23} = +1, \quad \varepsilon_{13} = -1. \quad (32)$$

Then

$$\Omega_{123} = \varepsilon_{12}\varepsilon_{23}\varepsilon_{13} = -1. \quad (33)$$

The operator is

$$D_{\text{fr}} = dI + \eta k \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}. \quad (34)$$

Its spectrum is

$$\text{spec}(D_{\text{fr}}) = \{d + \eta k, d + \eta k, d - 2\eta k\}. \quad (35)$$

Thus the frustrated cycle changes the spectral type. It is not just a relabeling of local signs.

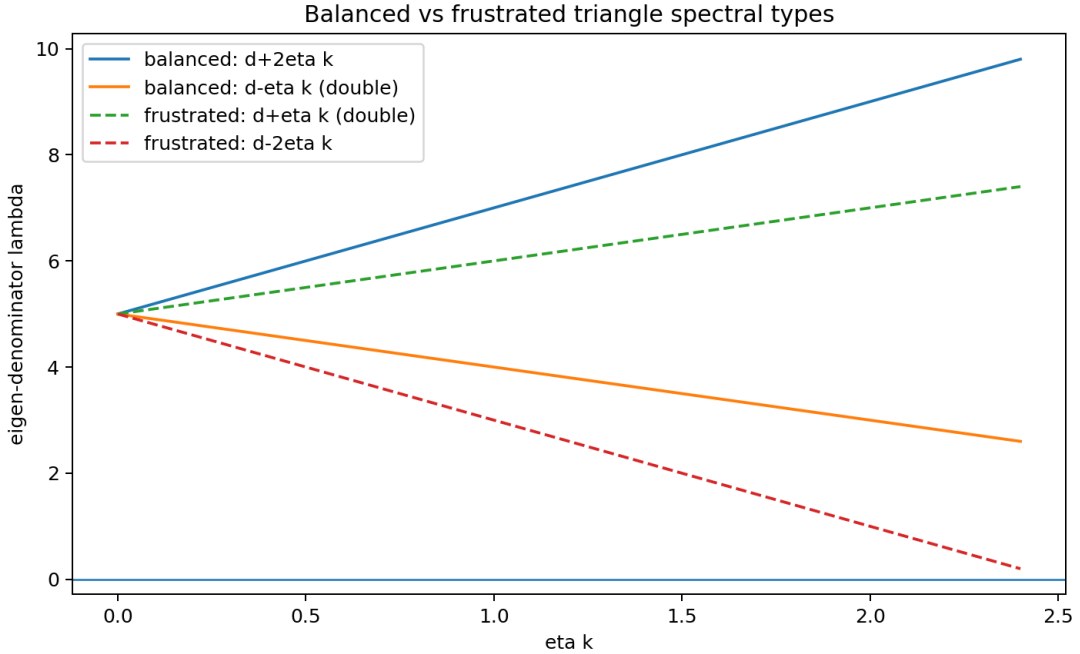


Figure 1: Balanced and frustrated triangle spectral types as the edge scale ηk is varied. The frustrated class has a different horizon margin: its lowest denominator is $d - 2\eta k$ rather than $d - \eta k$.

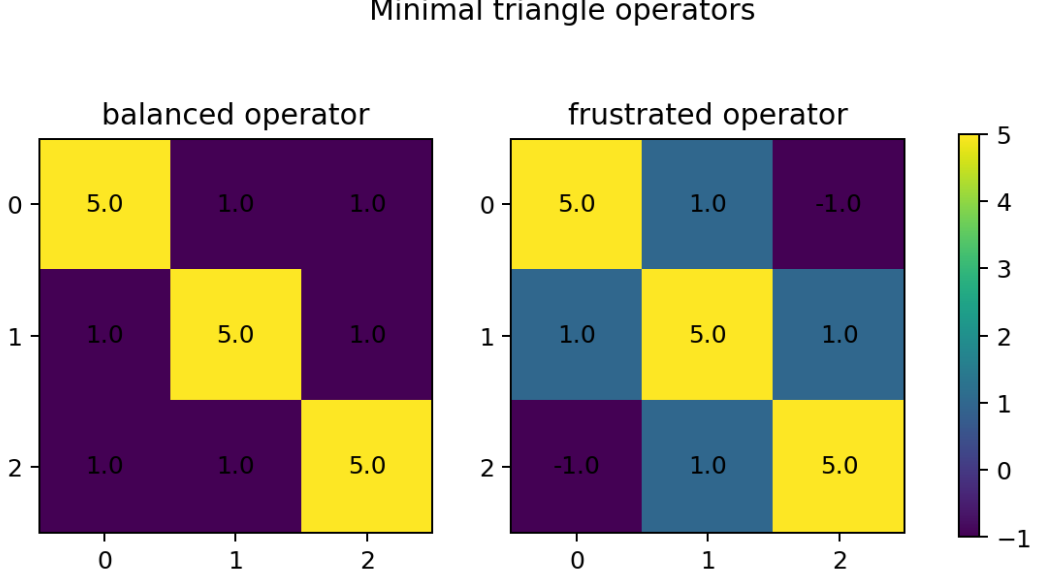


Figure 2: Minimal triangle operators. The frustrated class cannot be converted to the balanced class by vertex reorientation because the cycle product is -1 .

7 Admissibility horizon and localization price

BC-Origin I interpreted positive eigen-denominators through observable scale ratios

$$S_a = \frac{\mu_{\text{eff}}}{\lambda_a}, \quad \lambda_a > 0. \quad (36)$$

Therefore spectral splitting is not merely a drawing of lines on a graph. It changes the localization scale of observable shadows.

The structural gap between eigen-denominators can be read as a finite-dimensional localization price: hidden coupling prevents exact degeneracy, separates the spectral branches, and thereby compresses one branch while pushing another toward the horizon.

For the minimal two-shadow model

$$D(t) = \begin{pmatrix} d_1(t) & \kappa \\ \kappa & d_2(t) \end{pmatrix}, \quad (37)$$

the eigen-denominators are

$$\lambda_{\pm}(t) = \frac{d_1(t) + d_2(t)}{2} \pm \sqrt{\left(\frac{d_1(t) - d_2(t)}{2}\right)^2 + \kappa^2}. \quad (38)$$

If $\kappa \neq 0$, the branches avoid crossing. This is Landau-Zener-type avoided-crossing geometry in a finite-dimensional toy operator. It is not, by itself, a Landau-Zener transition-probability model and it is not a physical time-evolution law.

In BC-Origin III this observation is used only as interpretation. The main theorem-level content remains the graph-holonomy classification and its spectral consequences.

8 Relation to the Flow Lab

The software companion contains two layers.

The first layer is the BC-Origin III holonomy calculation package. It generates the triangle spectra, cycle-product diagnostics and heatmaps used in this manuscript.

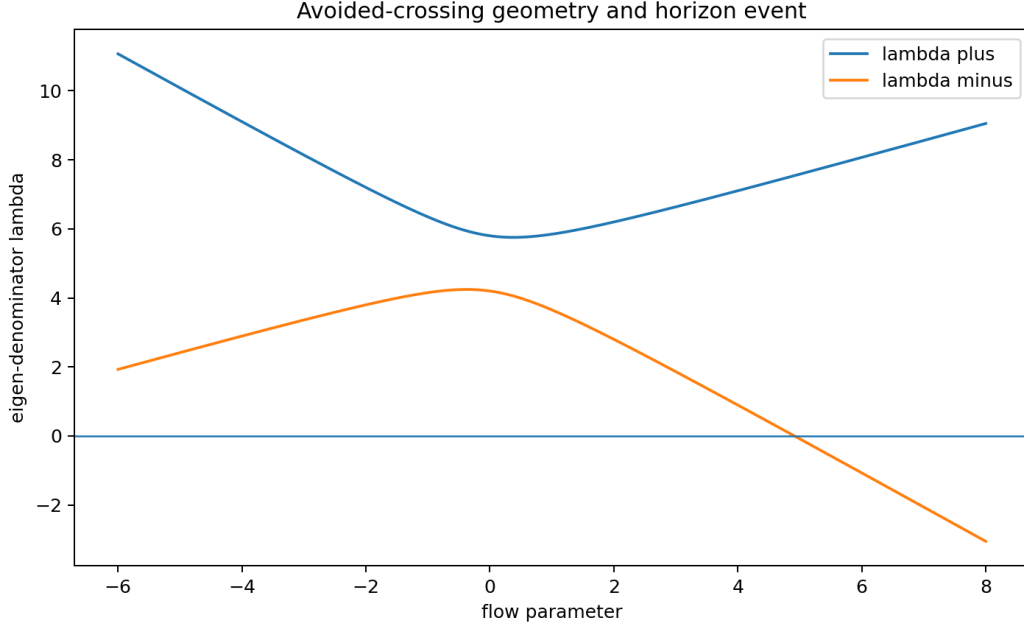


Figure 3: Companion flow-lab intuition. A nonzero structural coupling creates avoided-crossing geometry. The lower branch may approach the admissibility horizon $\lambda = 0$; this is a toy-model horizon event, not a claim of physical annihilation.

The second layer is an optional BC-Origin Flow Lab. It visualizes a two-shadow parameter-flow toy model with avoided crossing and horizon crossing. This Flow Lab is a preview direction for a later BC-Origin IV article. The flow parameter should be read as a controlled deformation parameter, not automatically as physical time.

This separation prevents the present paper from mixing two claims:

1. BC-Origin III: non-factorizable edge holonomy creates genuine cycle obstruction.
2. BC-Origin Flow/IV: parameter-dependent shadow operators can show avoided crossings and horizon events.

The first claim is the article’s formal contribution. The second is a visual and conceptual companion.

9 Controlled claims

The paper claims:

1. The baseline BC-Origin II sign rule is balanced because it is vertex-factorizable.
2. A non-factorizable edge-holonomy field ε_{ij} is the minimal $\mathbb{Z}_2$ graph extension required for frustrated cycle products.
3. Cycle products Ω_C are invariant under vertex reorientation.
4. A connected graph sign field is vertex-factorizable if and only if every cycle product is $+1$.
5. The holonomy-weighted off-diagonal shadow operator has gauge-invariant spectral classes under vertex sign changes.
6. The minimal triangle has different balanced and frustrated spectral types.
7. These spectral differences change toy-model admissibility-horizon margins.

10 Non-claims and limitations

The paper does not claim:

1. that ε_{ij} is a real physical gauge field;
2. that the model is a lattice gauge theory;
3. that it derives real spin-glass physics;
4. that spectral gaps are particle masses;
5. that horizon crossing is physical annihilation;
6. that the Flow Lab defines physical time;
7. that the model has empirical validation;
8. that it reconstructs a true hidden sector.

The entire construction remains a finite-dimensional structural-existence model.

11 Software appendix

The accompanying software package has the following suggested structure:

```
bc_origin/origin_iii/  
  README.md  
  software/  
    bc_origin_iii_holonomy_core.py  
    generate_bc_origin_iii_figures.py  
  figures/  
    triangle_holonomy_spectra.png  
    balanced_vs_frustrated_heatmaps.png  
    cycle_holonomy_product.png  
    flow_avoided_crossing_horizon.png
```

```
bc_origin/origin_flow/  
  README.md  
  web/  
    index.html
```

The holonomy figures are generated by

```
python software/generate_bc_origin_iii_figures.py --out figures
```

from the `bc_origin/origin_iii` folder.

The Flow Lab is a standalone browser file and can be opened directly:

```
bc_origin/origin_flow/web/index.html
```

12 Conclusion

BC-Origin I showed that a hidden oriented residual generator can produce a toy observable shadow with scale, sign and local admissibility. BC-Origin II replaced pairwise fitted coupling by structural kernels on hidden winding-index space. BC-Origin III adds the next minimal layer: non-factorizable edge holonomy.

The main result is conceptual and finite-dimensional. If all signs are products of vertex labels, the graph is balanced and frustration is absent. If an edge-holonomy field has a negative cycle product, no vertex reorientation can remove it. This non-removable cycle data changes the spectral class of the multi-shadow operator and therefore changes the localization margins of the observable toy shadows.

The programmatic role of BC-Origin III is to open a controlled path from structural kernels to frustrated spectral geometry, while keeping the stronger dynamical interpretation for the separate Flow Lab and a later BC-Origin IV module.

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